

National University of Computer and Emerging Sciences

Information Retrieval (CS4051)

Date: May 20, 2026 (12:30PM-03:30PM)

Course Instructors

Muhammad Rafi

Final Examination

Total Time: 3 Hour

Total Marks: 100

Total Questions: 07

Semester: Spring-2026

Campus: Karachi

Department: AI & DS

Student Name

Roll No

Section

Student Signature

CLO # 1: <Understand the basic concepts and techniques in Information Retrieval.>

Logical and Conceptual questions from chapter 1-7

Question No. 1

[Time: 20 Min] [Marks: 10]

- a. In practical Implementation of Vector Space Model (VSM), what is the major problem you have observed? illustrate.

The major problem is high dimensionality and sparsity. In VSM, every unique term becomes a dimension. For a large document collection, vocabulary may contain thousands or millions of terms, but each document uses only a small subset. This causes: large memory usage, slow similarity computation, noisy features, difficulty in handling synonyms and semantic meaning

- b. How context sensitive spelling corrections are performed in modern search engines?

Modern search engines perform spelling correction by using both word form and context.

For example: Query: "I want to by a book" ⇔ Correction: "I want to buy a book"

Here, "by" is a valid word, but in this context, "buy" is more appropriate. Search engines use: query logs, n-gram language models, word embeddings/contextual embeddings, edit distance, user click behavior, transformer-based models. They choose the correction that is most probable in the given context.

- c. In the context of VSM, suggestion some of the improvements in computing the following

$$\text{Cosine Similarity}(d, q) = \frac{d \cdot q}{\|d\| \|q\|}$$

The computation of the above cosine similarity is quite expensive to reduce its cost we can apply the following mathematical changes to it. For both d and q use unit vectors hence

National University of Computer and Emerging Sciences

the denominator is removed from this, the second manipulation is in the rank ordering as $\cos(\Theta) = \Theta$ for the Θ ranges from 0 to 180 degree. Hence we only left with the dot product of d and q . This only requires to multiple the common terms in document and query.

- d. Consider the wild card query of the form lo^*tion , assuming a bi=gram index is used to fetch the result-set. Give two terms one as true positive and one as false negative search result?

The query lo^*tion means terms starting with lo and ending with $tion$. Possible bi-grams include ($\$l, lo, ti, io, on, n\$$)

True positive term: location Because it matches the pattern lo^*tion . False positive term: locomotion It may be retrieved by bi-grams, but it does not exactly match lo^*tion if the required ending is strictly “tion” after wildcard matching depending on filtering.

- e. What will be a range for values for a term frequency? Explain.

Term frequency means the number of times a term appears in a document. It ranges in the interval of $0 \leq tf(t,d) \leq |d|$ where $|d|$ is the number of token in the longest document.

- f. Indicate True/False- justify in one-liner. “Tokenization is a language dependent process.”

True. Different languages have different writing systems and word boundary rules, so tokenization depends on language.

- g. Indicate True/False- justify in one-liner. “Document frequency of a term can be zero”

True, in vocabulary/query context. If a query term does not appear in any document of the collection, its document frequency is zero. But for terms already selected from the collection vocabulary, document frequency is at least 1.

- h. Indicate True/False – Justify in one liner. “We can estimate number of terms in a collection.”

True. We can estimate the number of terms using corpus under Heaps law.

- i. Indicate True/False – Justify in one liner. “Champion list can be precomputed”

True. Champion lists store top documents for each term in advance, so they can be precomputed to speed up retrieval.

- j. Indicate True/False – Justify in one liner. “Positional index are good for phrase queries”

True. A positional index stores the positions of terms in documents, so it helps check whether words occur together in the correct order.

National University of Computer and Emerging Sciences

CLO # 4: <Appreciate the importance of data structures, such as an index, to allow efficient access to the information in large bodies of text, identify the storage and index requirements for different IR Models>

Logical and Conceptual Questions from chapter (8,9,11,12)

Question No. 2

[Time: 20 Min] [Marks: 10]

- a. Explain one problem associated with using cumulative gain as evaluation metrics for IR?

The cumulative gain is a simple sum of relevance of the document – it does not consider the positioning of the documents. Hence this will be same for any permuted order of the search results with the same documents.

- b. What are some good ideas in collecting indirect relevance feedback?

Indirect relevance feedback is collected by observing user behavior without directly asking whether a document is relevant or not. For example, if a user clicks a search result, spends more time on a page, scrolls deeply, bookmarks, shares, or downloads a document, the system may treat it as a sign of usefulness. Similarly, if the user quickly reformulates the query, it may indicate that the previous results were not satisfactory. These signals help the search engine improve ranking and understand user needs better.

- c. In Probability Ranking Principle (PRP), why there is an assumption for null document i.e. $P[d=0/R=1, q] = P[d=0/R=0, q] = 0$?

The null document assumption, assume that null documents are no relevance to relevant or non-relevant documents. The null document is not an actual document in the collection. It contains no information, so it should not affect ranking. In simple words, PRP ranks real documents only, not an empty or non-existing document.

- d. How dynamic search results snippets computed? Explain.

Dynamic search snippets are query dependent document extracts that shown below the search result(documents). This gives good idea to a user's on why the document is retrieved for the query. A search engine finds the query terms inside a document, selects the most useful text around those terms, highlights the important words, and may join different parts using ellipses. For example, for the query "retrieval models", the snippet may show the sentence portion where these words appear, helping the user quickly judge whether the document is relevant.

- e. List two drawbacks of language model for IR.

Zero probability problem -If a query term does not appear in a document, the document may get zero probability without smoothing.

Independence assumption - Basic language models assume query terms are independent, ignoring word order and deeper meaning.

CLO # 2: < Understand how statistical models of text can be used to solve problems in IR, with a focus on how the vector-space model and language models are implemented and applied to document retrieval problems.>

Numerical Problem from chapter (4,5,6,7)

Question No. 3
10+10]

[Time: 30 Min] [Marks:

- a. Consider the partial document collection $D = \{d1: w1 w2 w4 w1; d2: w3 w2 w6; d3: w1 w2 w7\}$ and $q: w4 w3 w7$; if the following table gives the **tf** and **idf** score of each term, compute the score of each document against the given query q (assume query use simple tf_q scores), using cosine of angle between query vector and document vector. Give vector representations of documents vector and query. Also produce the ranking of the documents against this query.

Word	tf-d1	tf-d2	tf-d3	idf
W1	0.34	0.17	0.12	0.14
W2	0.12	0.29	0.19	0.38
W3	0.23	0.33	0.14	0.51
W4	0.26	0.28	0.22	0.24
W5	0.15	0.66	0.15	0.60
W6	0.31	0.22	0.16	0.32
W7	0.23	0.45	0.21	0.15

$d1 = [0.0476, 0.0456, 0.1173, 0.0624, 0.0900, 0.0992, 0.0345]$

$d2 = [0.0238, 0.1102, 0.1683, 0.0672, 0.3960, 0.0704, 0.0675]$

$d3 = [0.0168, 0.0722, 0.0714, 0.0528, 0.0900, 0.0512, 0.0315]$

$q = [0, 0, 1, 1, 0, 0, 1]$

$\cos(d1, q) = 0.6098$

$\cos(d2, q) = 0.3800$

$\cos(d3, q) = 0.5675$

most relevant document against query - $d1 > d3 > d2$

- b. In an IR System there were 60 relevant documents for a given query "q". The system returned 108 documents in response to the same query. If 50% documents in the result-set are relevant, compute the Precision and Recall of the system?

Total relevant document in collection for query $q = 60$

Total returned document against query $q = 108$

50% documents are relevant in returned result = $50\% \times 108 = 54$

Hence :

Precision = $54/108 = 0.5$

Recall is $54/60 = 0.9$

National University of Computer and Emerging Sciences

CLO # 2: <Understand how statistical models of text can be used to solve problems in IR, with a focus on how the vector-space model and language models are implemented and applied to document retrieval problems>

Numerical Problem from chapter (8,9)	
Question No. 4	[Time: 20 Min] [Marks: 10]

The following list of Rs and Ns represents relevant (R) and non-relevant (N) returned documents in a ranked list of 20 documents retrieved in response to a query from a collection of 10,000 documents. The top of the ranked list (the document the system thinks is most likely to be relevant) is on the left of the list. This list shows 6 relevant documents. Assume that there are 8 relevant documents in total in the collection.

R R N N N N N N R N R N N N R N N N N R

(1) What is the precision of the system on the top 10?

We have 3 relevant documents in top 10 results so, $\text{precision@10} = 3/10 = 30\%$

(2) What is the recall of the systems on the top 20?

From the 8 total relevant documents, the top 20 list retrieved 6 so the $\text{recall@20} = 6/8 = 75\%$

(3) What is the F1 on the top 20?

We know $\text{precision@20} = 6/20 = 0.30$ and $\text{Recall@20} = 0.75$ Now, F1 is the harmonic mean of Precision and Recall so $F1 = 2 \times 0.3 \times 0.75 / (0.3 + 0.75) = 0.4286 = 42.86\%$

(4) What is the largest possible MAP that this system could have?

For the largest possible MAP is possible when the remaining relevant documents are found at 21st and 22nd ranked.

$$\text{MAP}_{(\text{largest})} = 1/8 \times (1 + 1 + 3/9 + 4/11 + 5/15 + 6/20 + 7/21 + 8/22) = 50.80\%$$

(5) What is the smallest possible MAP that this system could have?

Similarly, the smallest possible MAP is when the remaining relevant documents were at the last retrieved result.

$$\text{MAP}_{(\text{smallest})} = 1/8 \times (1 + 1 + 3/9 + 4/11 + 5/15 + 6/20 + 7/9999 + 8/10000) = 41.64\%$$

CLO # 3: < Understand how statistical models of text can be used for other IR applications, for example clustering, classification and extraction of information>

Numerical Problem from chapter (11 and 12)	
Question No. 5 10+10]	[Time: 30 Min] [Marks:

- a. Consider the following document collection:

d1: w1 w3 w4 w5 w6 d2: w2 w3 w7 d3: w1 w2 w3 w4

Using the uni-gram language model, find the possible ranking of all the given documents for the query **Q: w4 w5**, you can use the following formula for the probability calculation
 $p(w_i) = (\text{count}(w_i) + 0.5) / (dN + 1)$. [10]

Using a uni-gram language model, we can have

$$P(Q/d1) = P(w4/d1) \times P(w5/d1) = (1+0.5)/(5+1) \times (1+0.5)/(5+1) = 0.0625$$

$$P(Q/d2) = P(w4/d2) \times P(w5/d2) = 0.5/4 \times 0.5/4 = 0.0156$$

$$P(Q/d3) = P(w4/d3) \times P(w5/d3) = 0.3 \times 0.1 = 0.03$$

Final Ranking $d1 > d3 > d2$

- b. Consider the following two documents

d1: w1 w3 w4 w5 w6 d2: w2 w3 w7

Let $\lambda = 1/2$ and Suppose the query is: “**w1 w2 w3**” estimate the following:

- Collection Probabilities for the query terms.
- MLE document probabilities with document model (M_d)
- Query likelihood estimate for the given document using mix mode with given λ

The formula is given as below:

$$\hat{P}(t|d) = \lambda \hat{P}_{mle}(t|M_d) + (1 - \lambda) \hat{P}_{mle}(t|M_c)$$

Where M_d and M_c have the same meaning we discussed during the lectures.

- Collection Probabilities for the query terms.

Total words in collection $5+3 = 8$

$$P(w1) = 1/8 ; P(w2) = 1/8 ; P(w3) = 2/8$$

- MLE document probabilities with document model (M_d)

$$\text{For } d1: P(w1/Md1) = 1/5 ; P(w2/Md1) = 0/5=0 ; P(w3/Md1) = 1/5$$

$$\text{For } d2: P(w1/Md2) = 0/3=0 ; P(w2/Md2) = 1/3 ; P(w3/Md2) = 1/3$$

National University of Computer and Emerging Sciences

(iii) Query likelihood using mixed model for (d1)

$$P(q|d1)=0.1625 \times 0.0625 \times 0.225$$

$$P(q|d1)=0.002285$$

Now for d2

$$P(q|d2)=0.0625 \times 0.229 \times 0.292$$

$$P(q|d2)=0.00418$$

Hence d2 is more relevant to the query q than d1.

National University of Computer and Emerging Sciences

CLO # 3: <Understand how statistical models of text can be used for other IR applications, for example clustering, classification and extraction of information.>

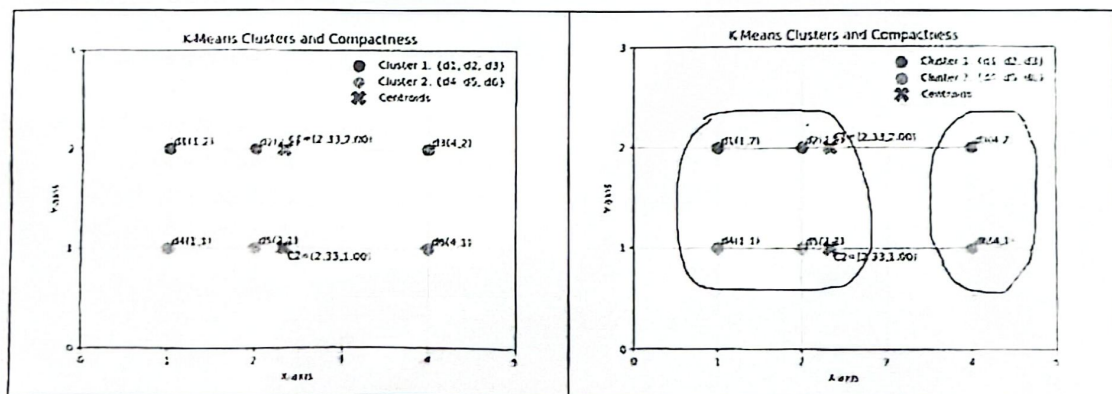
Logical and Conceptual problems on Neural IR, Word Embedding's, Classification, Clustering and Link	
Question No. 6	[Time: 20 Min] [Marks: 5+5]

- a. What are dense vectors in information representation? Give an example.

Dense vectors are low-dimensional continuous vector representations where most values are non-zero. These vectors can be learned using neural networks. They are commonly used to represent words, documents, or images in machine learning/NLP. Consider an example, if we want to represent a word "king" - we can use words like [king, queen, man and woman] to represents king like a vector of [king queen man woman] = [0.91, 0.56, 0.97, -0.67] Unlike sparse vectors such as one-hot vectors, dense vectors capture semantic meaning. For example. vectors of king, queen, man, and woman may be close in semantic space.

- b. Consider the following set of data points $D = \{d1(1,2), d2(2,2), d3(4,2), d4(1,1), d5(2,1), d6(4,1)\}$ taking d2 and d5 what will be the final clusters using K-Means. Is this solution optimal?

With initial seeds $C1 = (2,2)$ and $C2 = (2,1)$, we will get two clusters $Cluster1 = \{d1, d2, d3\}$ and $Cluster2 = \{d4, d5, d6\}$, while plotting we can clearly see that a more compact clusters would be $Cluster1 = \{d1, d2, d4, d5\}$ and $Cluster2 = \{d3, d6\}$



CLO # 4: <Understand the issues involved in providing an IR service on a web scale, interactive and neural information retrieval>

Numerical problems on Neural IR, Word Embedding's, Classification, Clustering and Link Analysis	
Question No. 7	[Time: 30 Min] [Marks: 20]

Q7. Answer the following questions.

- a. Explain the working of HITS algorithm - by defining Hub/Authority pages and iterative computation. [5]

Authority Page -An authority page contains valuable information on a topic and is pointed to by many good hub pages.

Hub Page - A hub page contains links to many useful authority pages.

HITS- Algorithm:

Iterative Process

Initialize all scores to 1 (Hub and Authority initialize to some random numbers)

Update authority scores (using Hub scores)

Update hub scores (using Authority score)

Normalize scores

Repeat until convergence

- b. Outline three differences between **HITS** and **PageRank**. [5]

HITS	PageRank
Computes two scores: Hub and Authority	Computes single importance score $pr(p_i)$
Query-dependent (computed at search time for a topic)	Query-independent (precomputed offline)
Works on a smaller subgraph related to the query	Uses the entire web graph
More sensitive to topic communities and link spam	More globally stable (Favor old pages)
Uses mutual reinforcement between hubs and authorities	Uses random surfer probability model

- c. Consider a specific subset of the web with four pages (p_1, p_2, p_3, p_4). The link structure is defined by the following adjacency matrix A given below:

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Where a and h define as:

$$a = A^T h \quad h = Aa$$

$$h_0 = [1, 1, 1, 1]^T \text{ and } a_0 = [1, 1, 1, 1]^T.$$

Using the above equation perform one complete iteration of HITS algorithm, give the normalized a and h vector. [10]

National University of Computer and Emerging Sciences

Consider the matrix A and the equations (i) $a_1 = A^T h_0$ and (ii) $h_1 = A \cdot a_1$

Consider the following:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad h^{(0)} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Compute Hub Scores:

$$h = Aa$$

Compute Authority Scores:

$$a = A^T h \quad ||a|| = \sqrt{0^2 + 1^2 + 2^2 + 2^2}$$

$$||a|| = \sqrt{9} = 3$$

$$a = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 2 \end{bmatrix} \quad a_{norm} = \begin{bmatrix} 0 \\ 0.333 \\ 0.667 \\ 0.667 \end{bmatrix}$$

$$||h|| = \sqrt{3^2 + 4^2 + 2^2 + 0^2}$$

$$||h|| = \sqrt{29} = 5.385$$

$$h_{norm} = \begin{bmatrix} 0.557 \\ 0.743 \\ 0.371 \\ 0 \end{bmatrix}$$

<The end.>